Print plan for the whole kit

For eleven children at three tables (K–1 four, grades 2–3 four, grades 4–5 three) and three adults. Print every sheet at 100% (Actual size), single-sided, on US Letter. Folding disks go on ordinary copy paper, which folds flat; card stock is too stiff to fold. Wedges, tabs and cards go on card stock. Check the 10 cm line on every sheet after printing.

Page	Sheet	Paper	K–1	2–3	4–5	Adults	Total	Used in
2	Disk, four right angles	copy	6	4	4	2	16	K–1 P1, P2, P3, P5; 2–3 P2 (A), P5; 4–5 P1 (A), P4, P5; launch
3	Disk B	copy		2	2		4	2–3 P2; 4–5 P1
4	Disk C	copy		2	2	1	5	2–3 P2; 4–5 P1; adult pretest
5	Disk D	copy		2	2	1	5	2–3 P2; 4–5 P1; adult pretest
6–9	Add-a-ray disks A, B, C, D	copy		2 each			8	2–3 P4
10	Blank disk, 15° rim ticks	copy	1	1	1	1	4	extra workspace
11	Seven lettered wedges	card		1			1	2–3 P1
12	Sixteen matching wedges	card		2			2	2–3 P3; 4–5 P3 (A)
13	Six wedges	card		1	1		2	2–3 P6; 4–5 P6, P3 (B)
14	K–1 M and V tabs	card	1				1	K–1 P4; recording
15	K–1 route cards A–H	card	1				1	K–1 P6

Totals. 42 sheets of copy paper and 7 of card stock. Nothing on these sheets reveals an answer, but cut the pieces before children arrive.

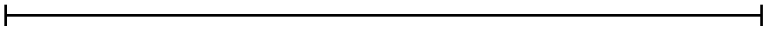
Disks. Cut on the circle, or trim each disk sheet to the 18 cm square on its corner marks with four straight cuts. Whether a disk folds flat is decided at its centre, so the square folds in exactly the same ways: any flat fold of the disk enlarges to the square, and any flat fold of the square trims down to the disk. Crease each printed ray both ways before class, pressing only along that ray from the centre out; a ray with no ray opposite it must not be continued through the centre, because that would add a ray. Precrease every disk this way before class. On pages 6–9 crease only the three printed rays: children choose, draw and crease the fourth. Replace a disk that gets an extra crease.

Wedges. Pieces in one set touch along shared cut lines, so each straight cut separates two pieces. Their radii match the student pages (2.25 cm, 2.5 cm, 2.4 cm), so a wedge can be laid on its printed picture. The half-turn boards and start disks stay whole. Keep each set in its own labelled envelope and never mix sets: the seven-piece and six-piece sets both use the letters A, B and C.

K–1. Cut the tabs apart (36 M and 36 V, four children's stocks for Problem 4) and the 16 route cards (two sets of A–H).

Time. About two and a quarter hours for one adult with scissors: 40 minutes of disk cutting (20 with squares and a paper trimmer), 45 of precreasing, 30 of wedges, 20 of tabs and cards.

Status. Digitally checked: every printed dimension and angle, and every crease pattern against the flat-folding conditions the problems need. Physically unrehearsed: no disk from this kit has been folded and no piece has been cut. The adult pretest in the guide (pp. 2–3) still comes first.

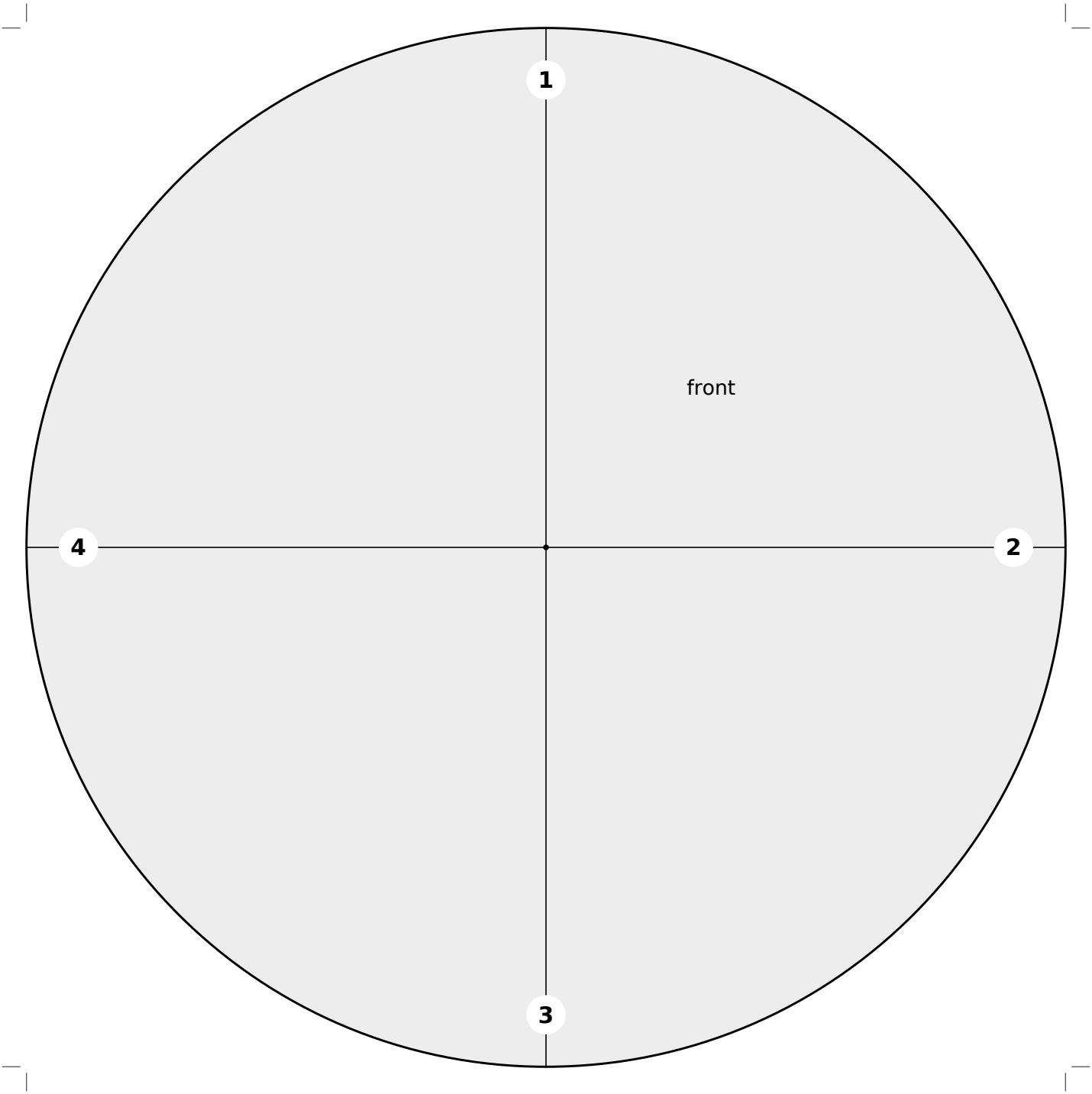


10 cm check: this line must measure 10 cm

Folding disk: four right angles

Print on copy paper, at 100%: K–1 6 · Grades 2–3 4 · Grades 4–5 4 · adults 2. K–1 Problems 1, 2, 3 and 5; grades 2–3 Problem 2 (disk A) and 5; grades 4–5 Problem 1 (disk A), 4 and 5; the launch. A spare uncut sheet makes a large recording disk.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease each printed ray both ways, pressing only along that ray from the centre out; make no other crease.

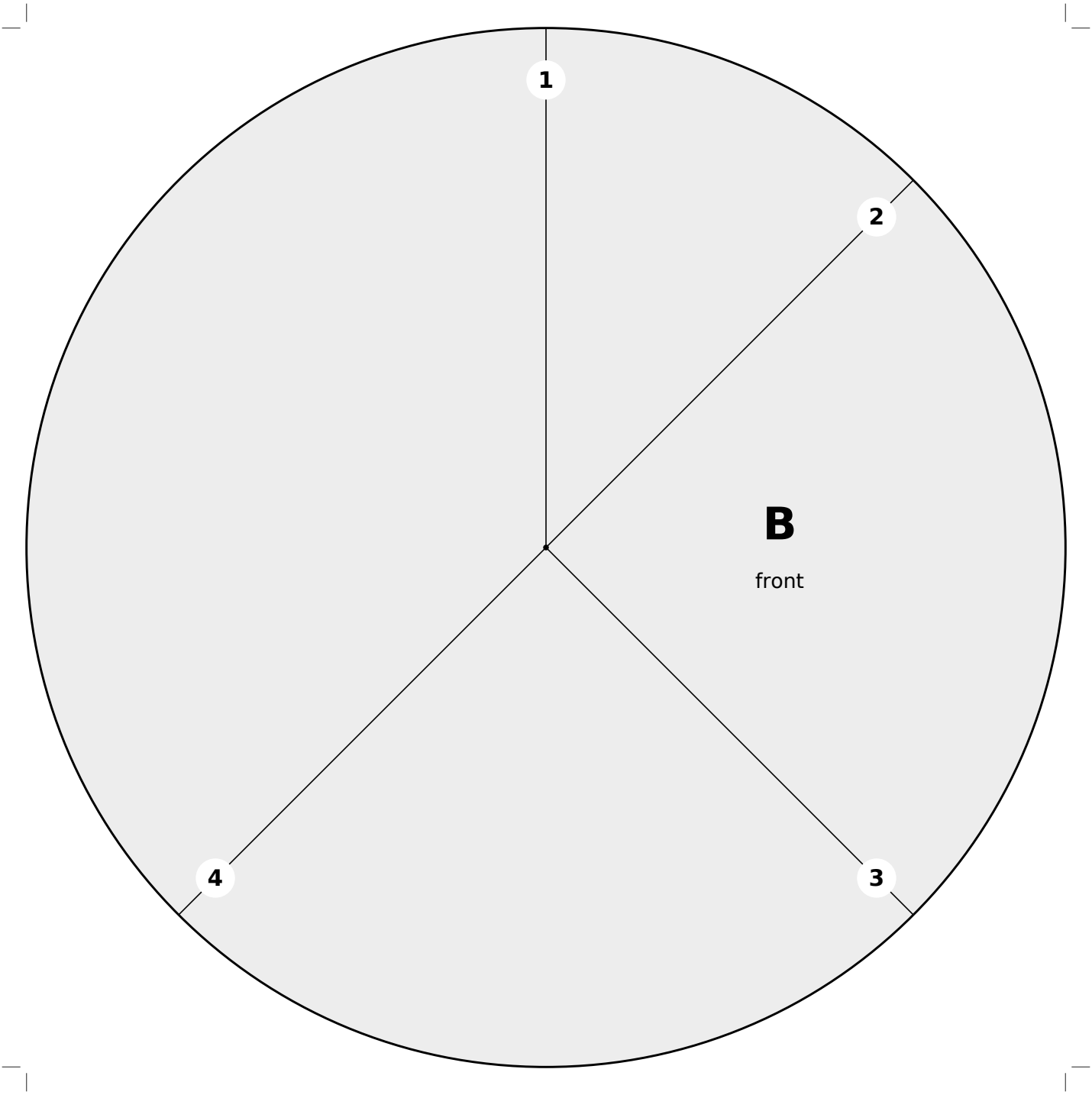


10 cm check: this line must measure 10 cm

Folding disk B: 45°, 90°, 90°, 135°

Print on copy paper, at 100%: Grades 2–3 2 · Grades 4–5 2. Grades 2–3 Problem 2 and grades 4–5 Problem 1, disk B.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease each printed ray both ways, pressing only along that ray from the centre out; make no other crease.

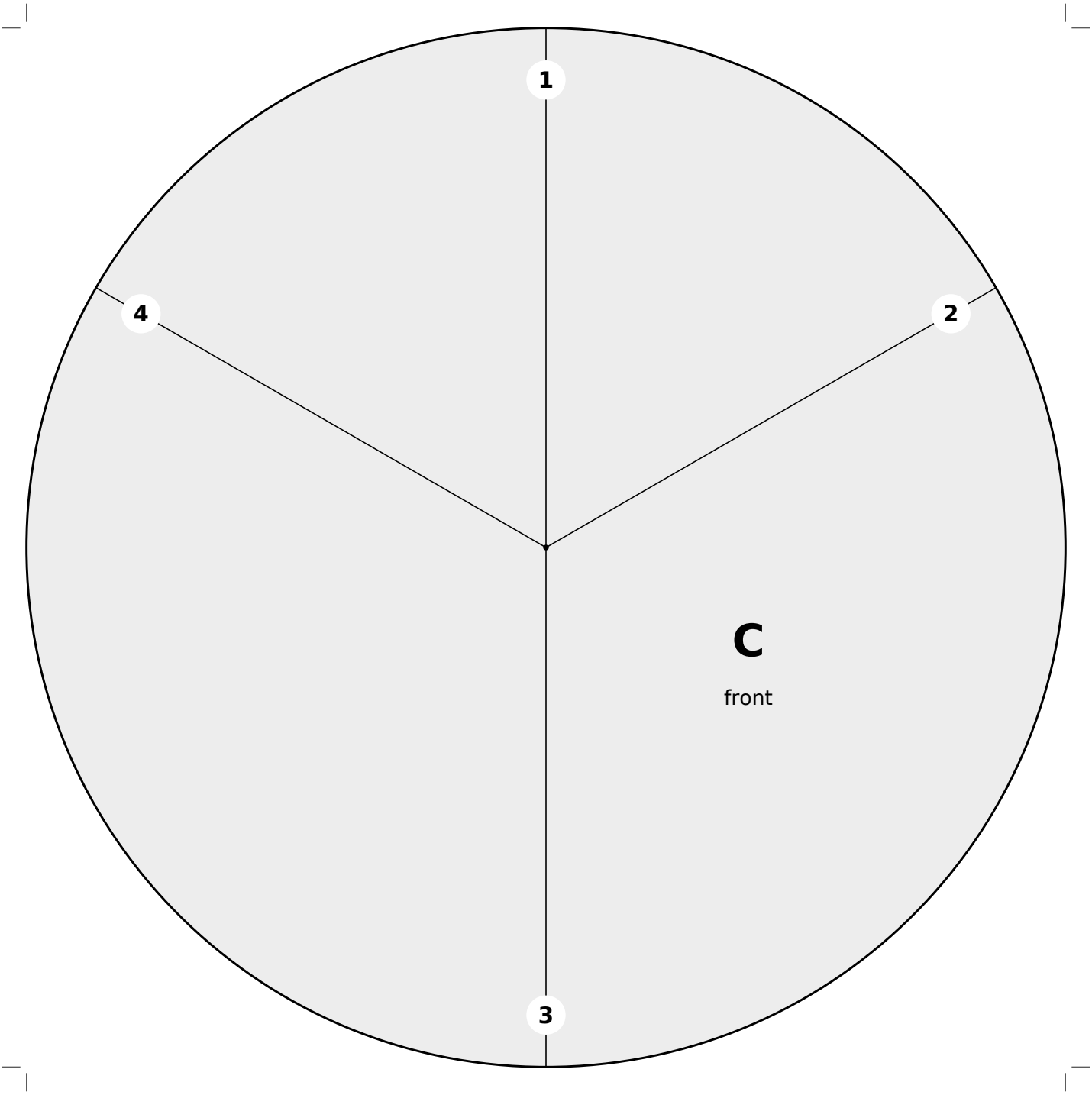


10 cm check: this line must measure 10 cm

Folding disk C: 60°, 120°, 120°, 60°

Print on copy paper, at 100%: Grades 2–3 2 · Grades 4–5 2 · adults 1. Grades 2–3 Problem 2 and grades 4–5 Problem 1, disk C. Adults: one for the pretest.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease each printed ray both ways, pressing only along that ray from the centre out; make no other crease.

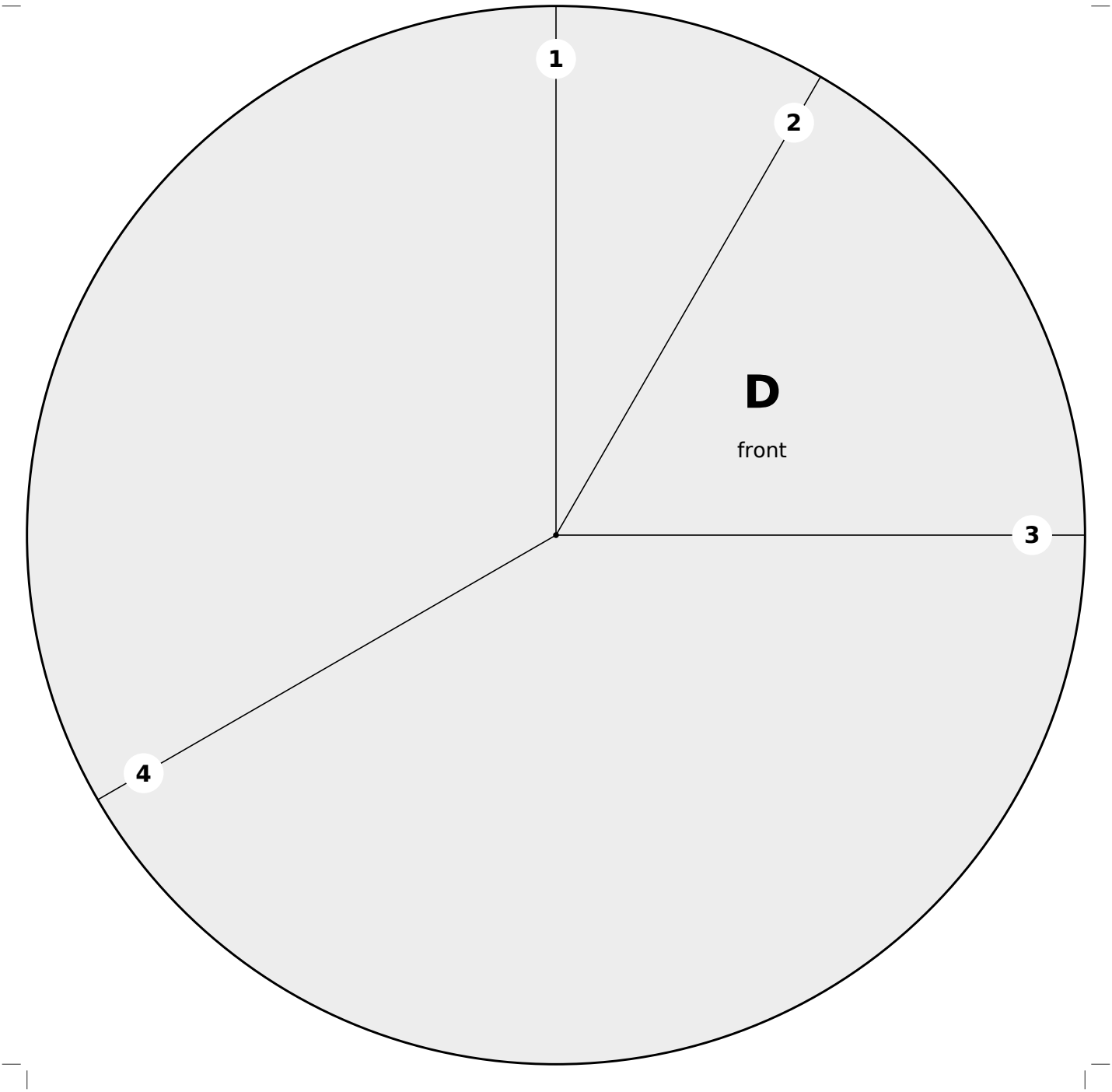


10 cm check: this line must measure 10 cm

Folding disk D: 30°, 60°, 150°, 120°

Print on copy paper, at 100%: Grades 2–3 2 · Grades 4–5 2 · adults 1. Grades 2–3 Problem 2 and grades 4–5 Problem 1, disk D. Adults: pretest its tuck (guide p. 3) and keep one successful fold as a spare.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease each printed ray both ways, pressing only along that ray from the centre out; make no other crease.

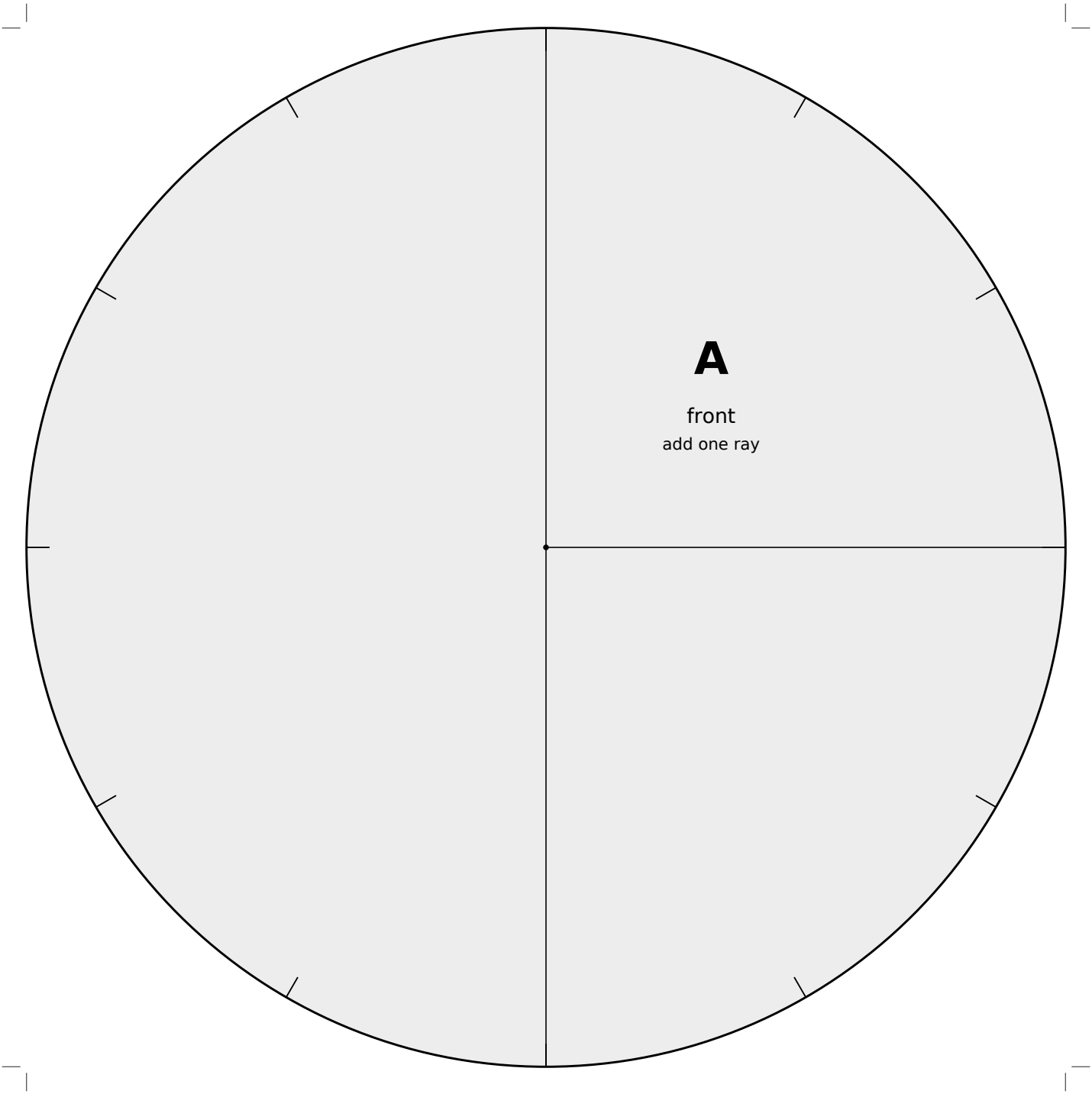


10 cm check: this line must measure 10 cm

Add-a-ray disk A: rays at 0°, 90° and 180°

Print on copy paper, at 100%: Grades 2–3 2. Grades 2–3 Problem 4, pattern A. Short ticks mark every 30° on the rim.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease only the three printed rays, both ways, pressing only along each ray from the centre out: a crease carried through the centre would add a ray. A child adds the fourth ray with a ruler, from the centre dot to a rim tick, then creases it.

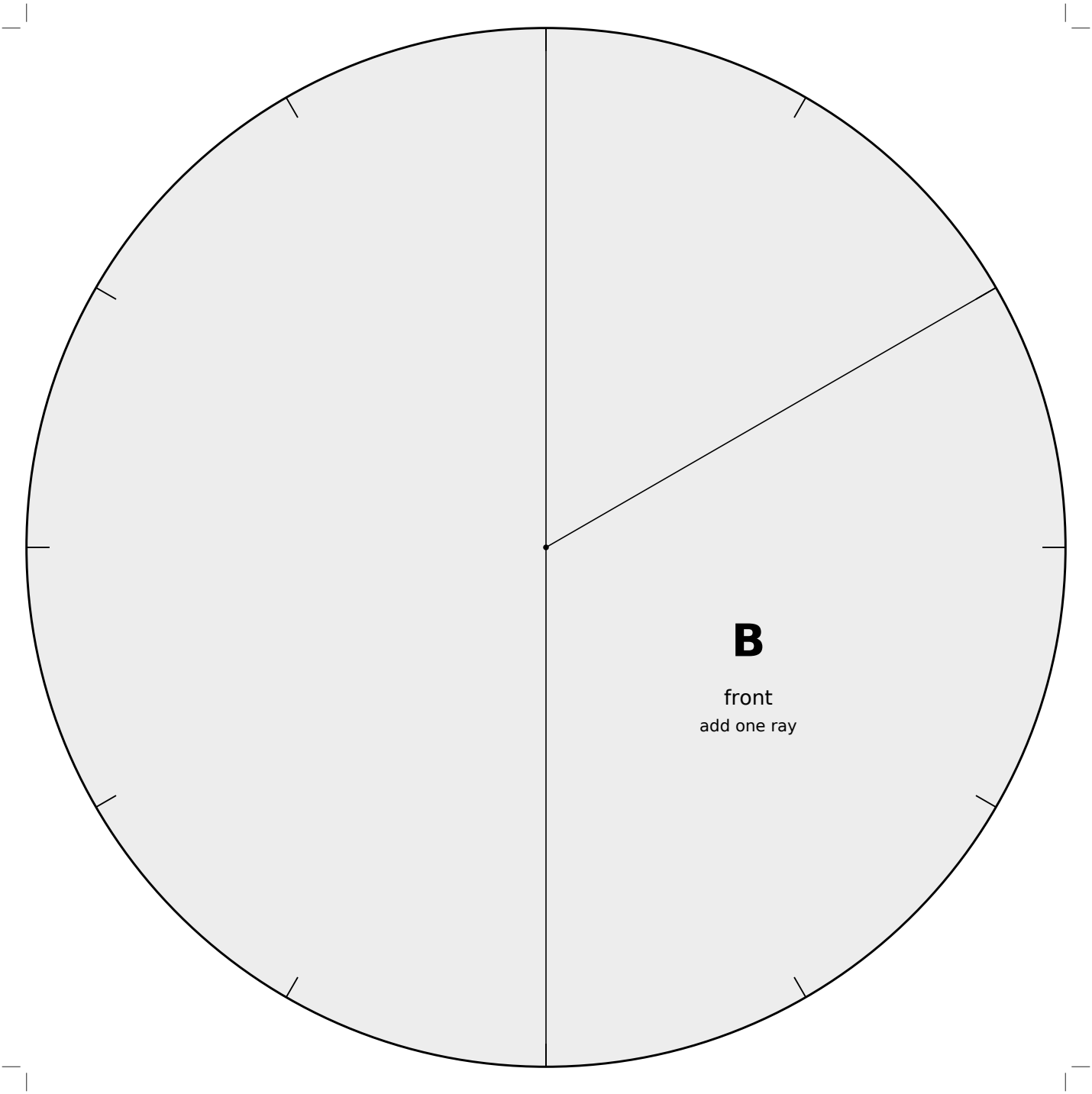


10 cm check: this line must measure 10 cm

Add-a-ray disk B: rays at 0°, 60° and 180°

Print on copy paper, at 100%: Grades 2–3 2. Grades 2–3 Problem 4, pattern B. Short ticks mark every 30° on the rim.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease only the three printed rays, both ways, pressing only along each ray from the centre out: a crease carried through the centre would add a ray. A child adds the fourth ray with a ruler, from the centre dot to a rim tick, then creases it.

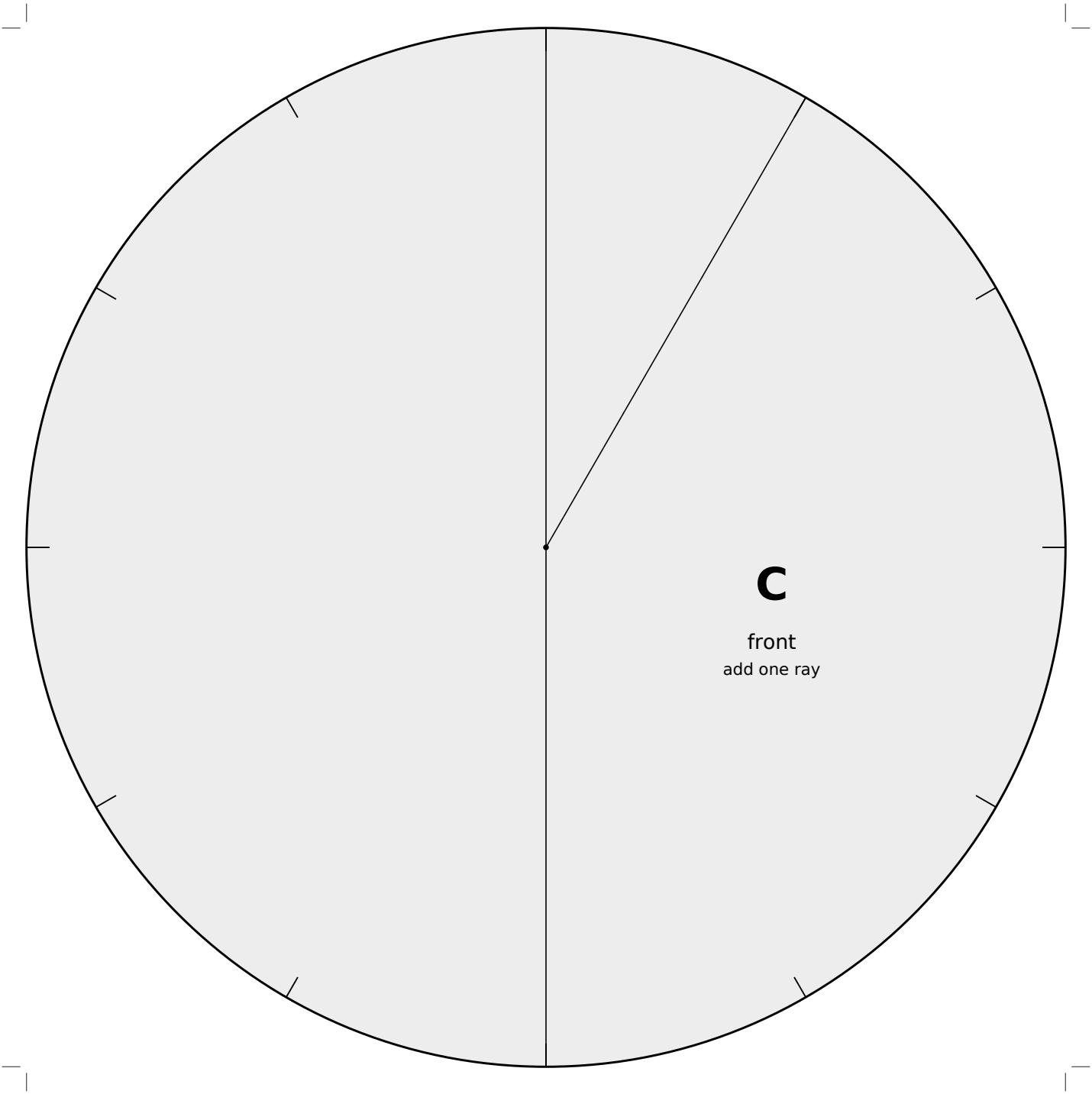


10 cm check: this line must measure 10 cm

Add-a-ray disk C: rays at 0°, 30° and 180°

Print on copy paper, at 100%: Grades 2–3 2. Grades 2–3 Problem 4, pattern C. Short ticks mark every 30° on the rim.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease only the three printed rays, both ways, pressing only along each ray from the centre out: a crease carried through the centre would add a ray. A child adds the fourth ray with a ruler, from the centre dot to a rim tick, then creases it.

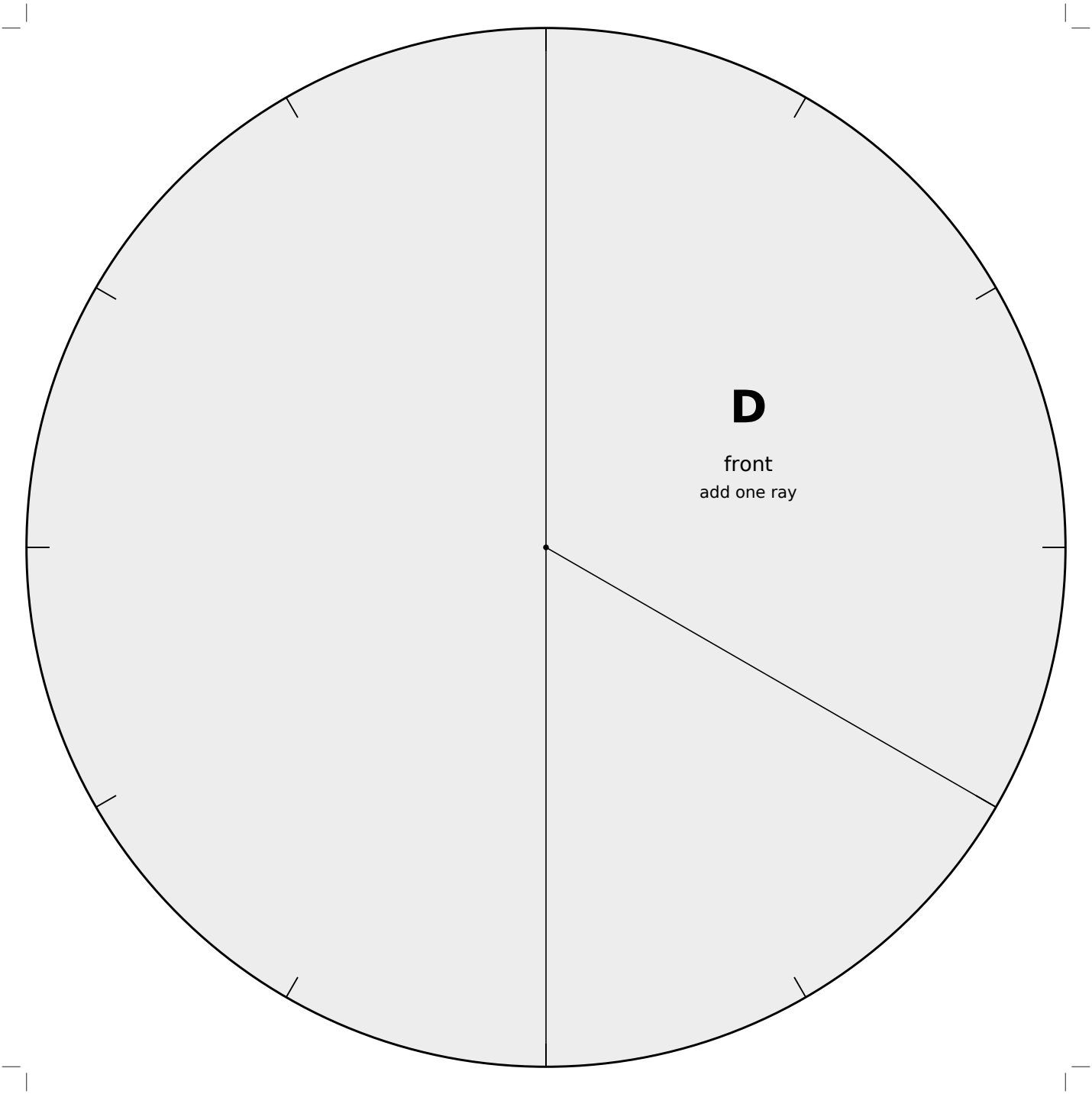


10 cm check: this line must measure 10 cm

Add-a-ray disk D: rays at 0°, 120° and 180°

Print on copy paper, at 100%: Grades 2–3 2. Grades 2–3 Problem 4, pattern D. Short ticks mark every 30° on the rim.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Crease only the three printed rays, both ways, pressing only along each ray from the centre out: a crease carried through the centre would add a ray. A child adds the fourth ray with a ruler, from the centre dot to a rim tick, then creases it.

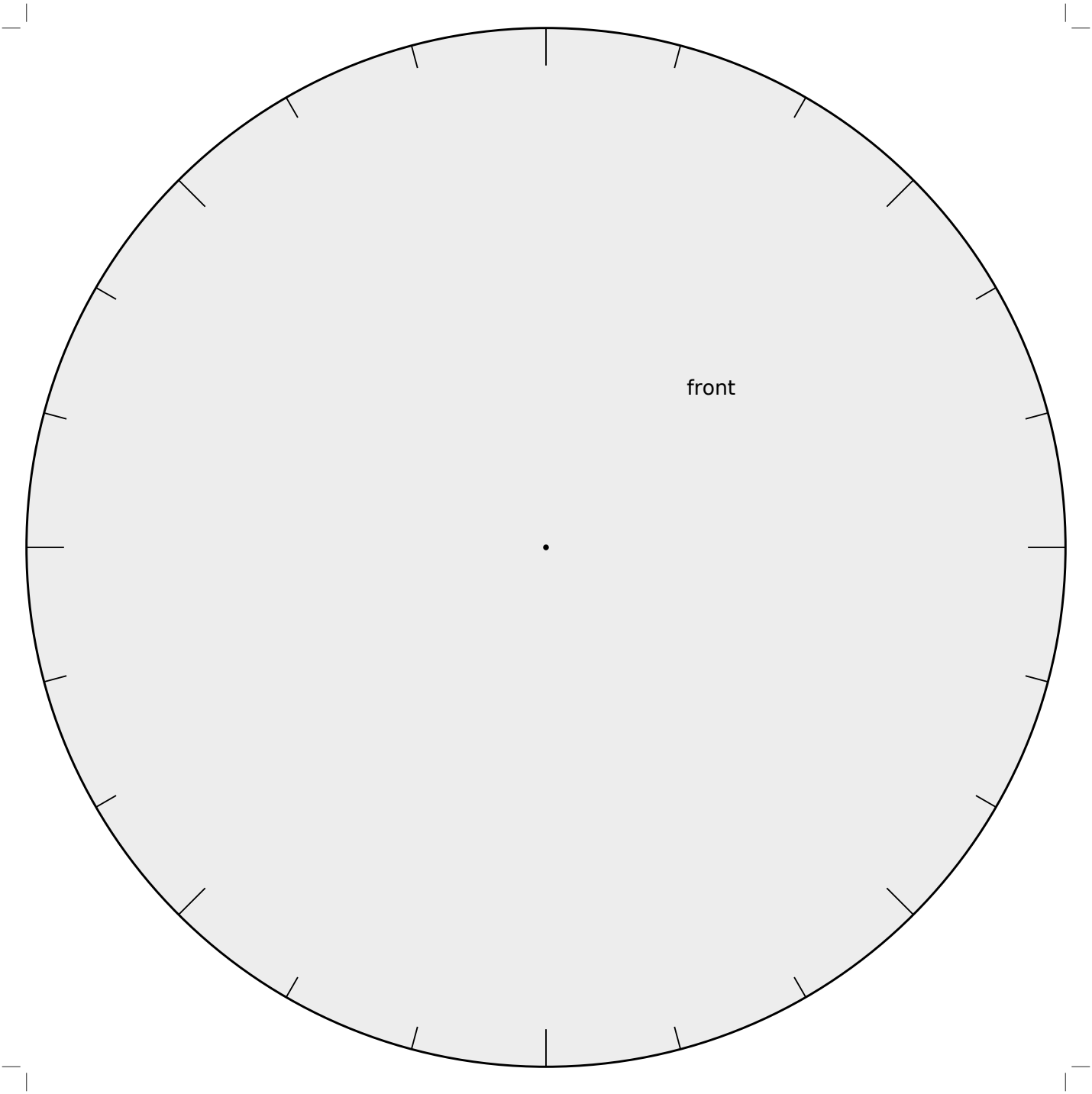


10 cm check: this line must measure 10 cm

Blank disk with 15° rim ticks

Print on copy paper, at 100%: K–1 1 · Grades 2–3 1 · Grades 4–5 1 · adults 1. Extra workspace: a pattern of a child's own, or a grades 4–5 pattern from Problems 2 and 3. Long ticks mark every 45°.

Cut on the circle, or trim to an 18 cm square on the four corner marks: a fold about one centre point behaves the same on both. The tinted side is the front. Draw rays with a ruler from the centre dot to the ticks before creasing.

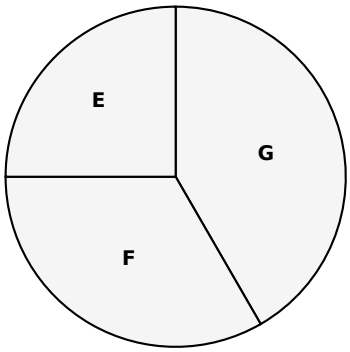


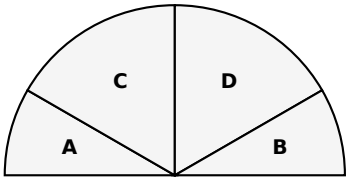
10 cm check: this line must measure 10 cm

Seven lettered wedges, grades 2–3 Problem 1

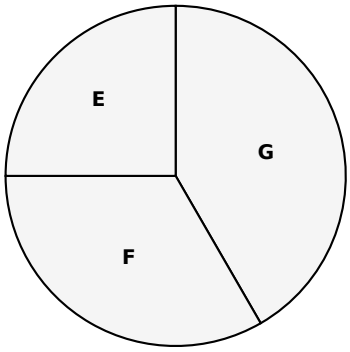
Print on card stock, at 100%: Grades 2–3 1. Three sets: one for each pair and a spare. A=30°, B=30°, C=60°, D=60°, E=90°, F=120°, G=150°, radius 2.25 cm, the size of the pictures and of the half-turn template on the Problem 1 page.

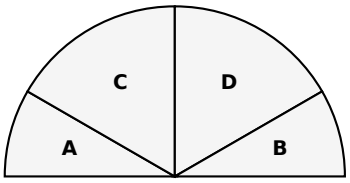
Cut each disk and half-disk on its outline, then along every line between letters. Put each set of seven in its own envelope.



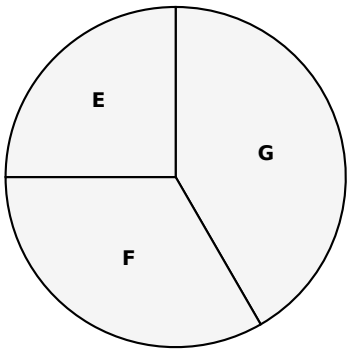


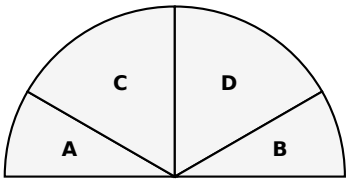
set 1



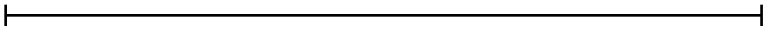


set 2





set 3

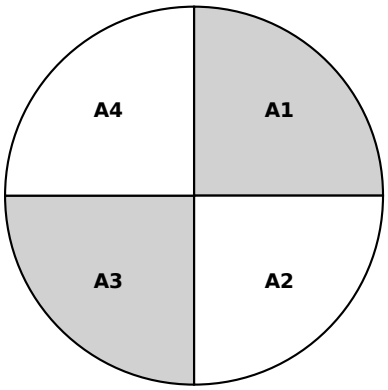


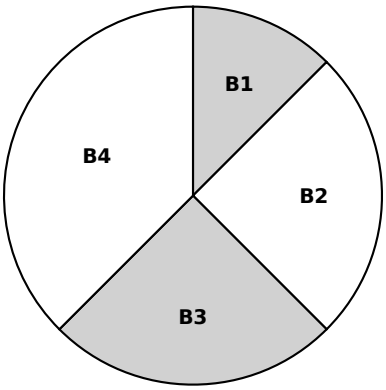
10 cm check: this line must measure 10 cm

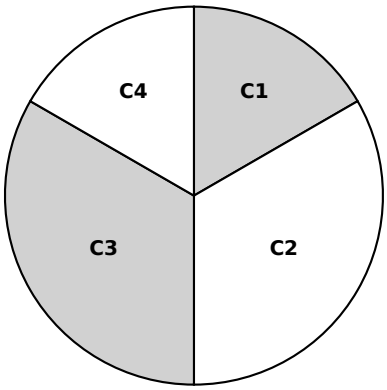
Sixteen matching wedges, grades 2–3 Problem 3

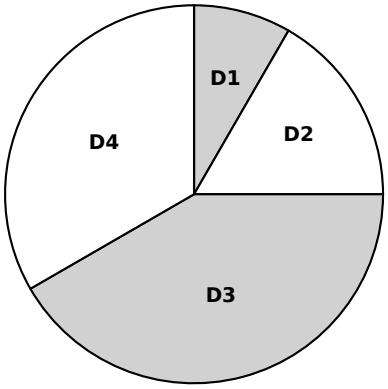
Print on card stock, at 100%: Grades 2–3 2. One set and two half-turn boards per sheet, so one sheet for each pair. Each disk is cut into the four sectors of the matching Problem 3 disk, radius 2.5 cm; sectors 1 and 3 are shaded as on the page. Disk B's pieces also give grades 4–5 Problem 3, pattern A.

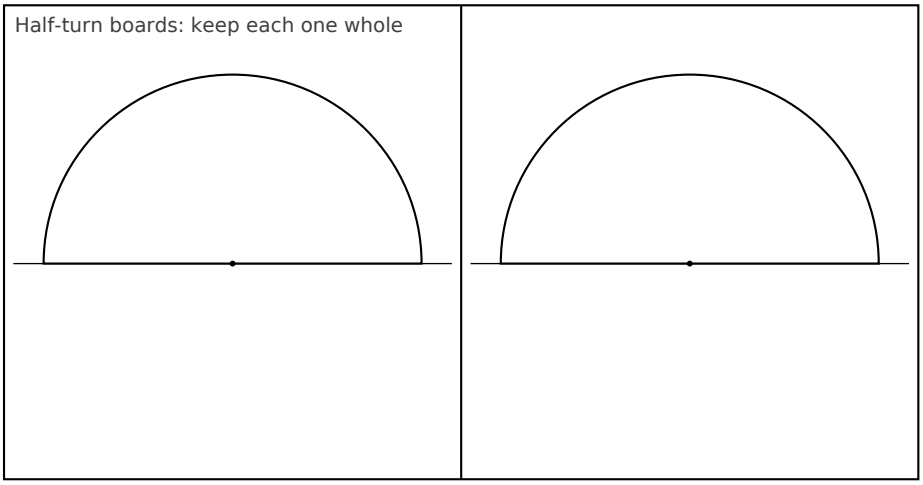
Cut each disk on its circle, then along its four rays to the centre. Keep the sixteen pieces in one envelope. Cut out the board strip whole, then into its two boards: one for the shaded pair and one for the white pair.









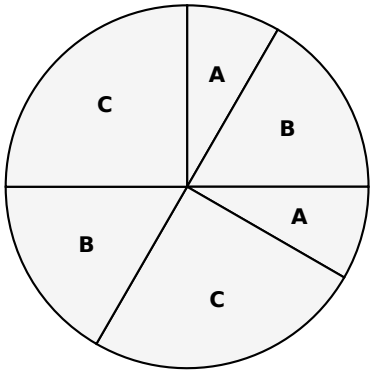


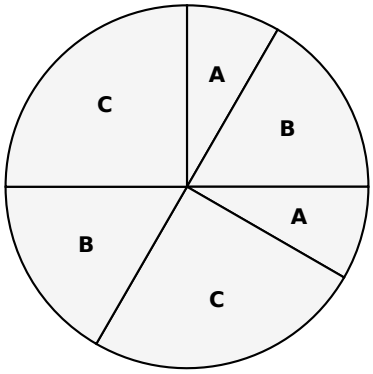
10 cm check: this line must measure 10 cm

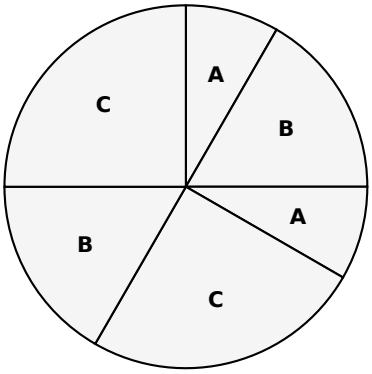
Six wedges, grades 2–3 and 4–5 Problem 6

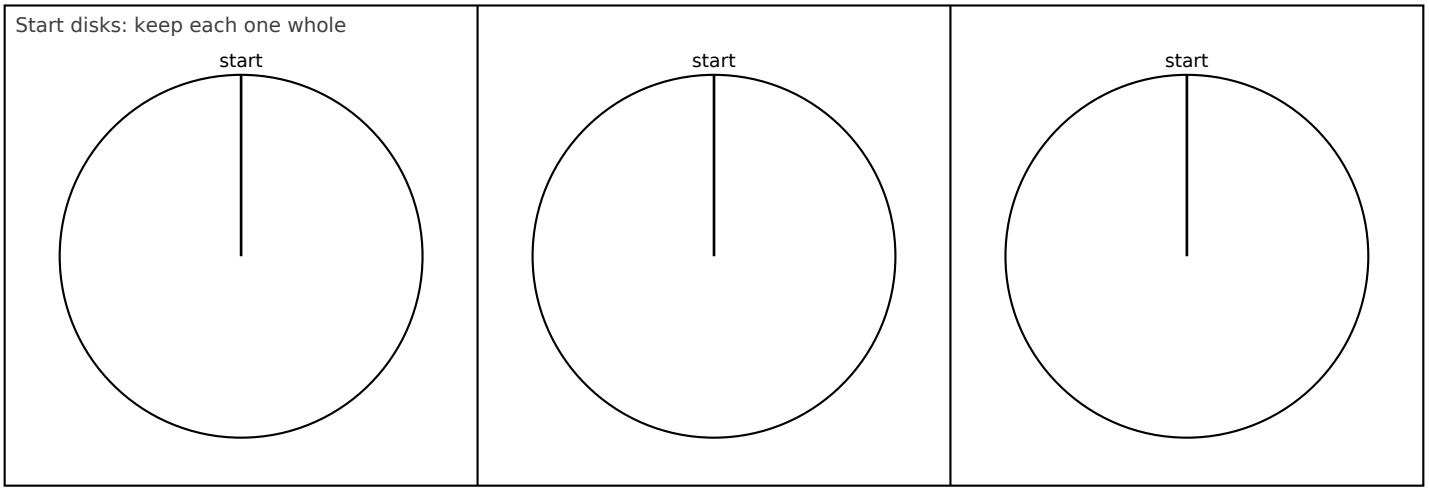
Print on card stock, at 100%: Grades 2–3 1 · Grades 4–5 1. Three sets: one for each pair and a spare. A, A = 30°, B, B = 60°, C, C = 90°, radius 2.4 cm as on the grades 2–3 page; equal letters are identical. The same set gives grades 4–5 Problem 3, pattern B.

Cut each disk on its circle, then along its six rays to the centre. Keep each set of six in its own envelope. The start disks stay whole: a board for filling a disk and reading clockwise from the top ray.









10 cm check: this line must measure 10 cm

K–1 M and V tabs, Problem 4

Print on card stock, at 100%: K–1 1. Four children's stocks of nine M and nine V tabs (each Problem 4 row takes the six its page shows), or spares for recording folds on Problems 1, 3 and 5. A printed stock box on the page is 9 mm; these tabs are 10 mm and cover it.

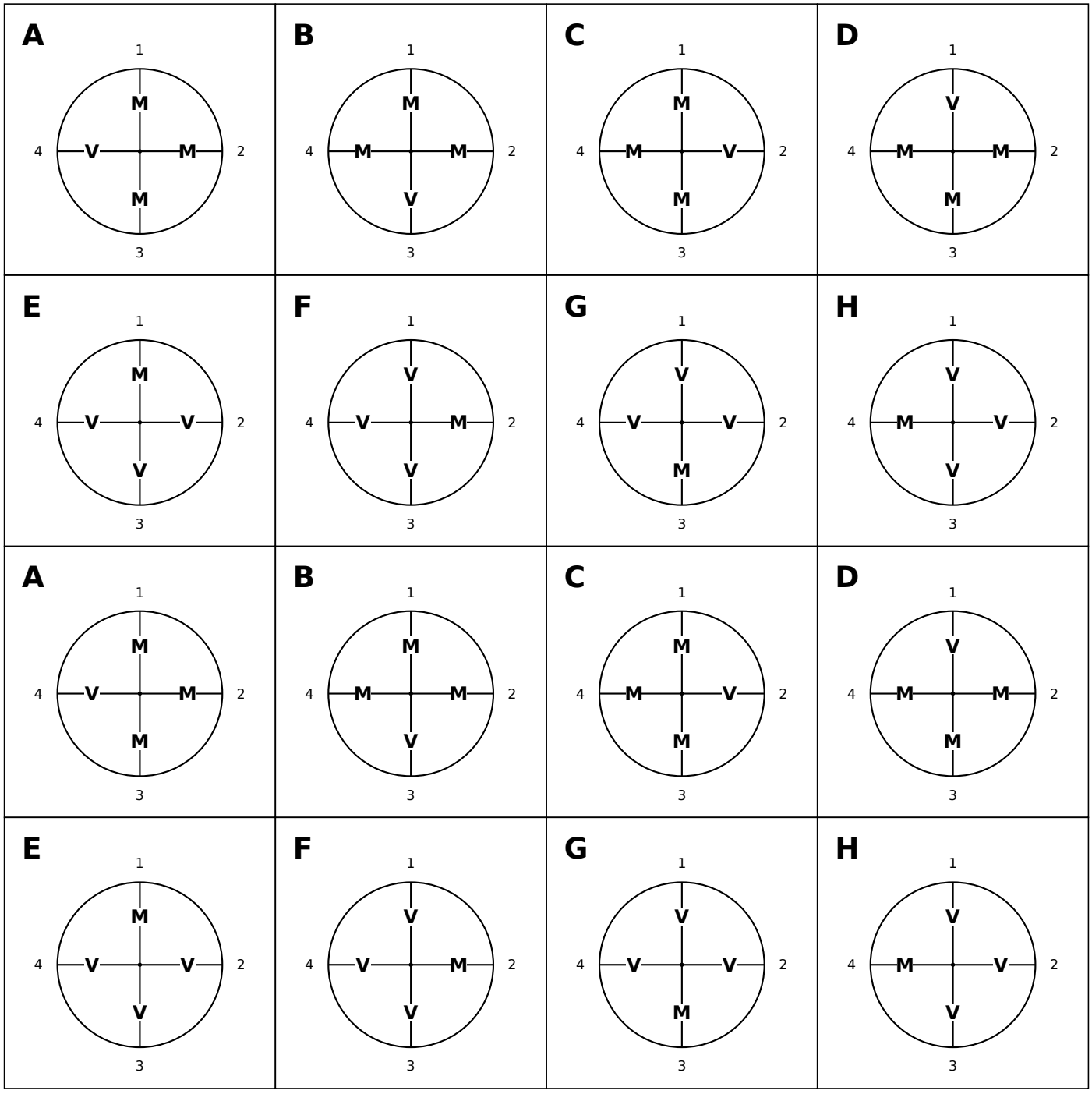
Cut along every line. Keep the tabs in an envelope.

M	**M**	**M**	**M**	**M**	**M**	**M**	**M**	**M**
M	**M**	**M**	**M**	**M**	**M**	**M**	**M**	**M**
M	**M**	**M**	**M**	**M**	**M**	**M**	**M**	**M**
M	**M**	**M**	**M**	**M**	**M**	**M**	**M**	**M**
V	**V**	**V**	**V**	**V**	**V**	**V**	**V**	**V**
V	**V**	**V**	**V**	**V**	**V**	**V**	**V**	**V**
V	**V**	**V**	**V**	**V**	**V**	**V**	**V**	**V**
V	**V**	**V**	**V**	**V**	**V**	**V**	**V**	**V**

10 cm check: this line must measure 10 cm

K–1 route cards A–H, Problem 6

Print on card stock, at 100%: K–1 1. Two sets, one for each pair. Each card is one of the eight folds printed on the Problem 6 page, with the same letter, so children can move cards into a route before writing it.
Cut along every line between cards.



10 cm check: this line must measure 10 cm